

DS-805 (Machine Learning System Engineering)

Credit Structure (L-T-P-Cr): 3 0 2 4

Content:

Unit 1 – ML Systems, Collaboration & Data Acquisition

- Overview of ML System Lifecycle and MLOps in production systems
 - Object-oriented Python for ML applications
 - Git/GitHub workflows: branching, merging, pull requests, code reviews
 - Web scraping & automation (Beautiful Soup, Scrapy)
 - Ethical and legal considerations in data acquisition
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Unit 2 – Containerization & Kubernetes for ML

- Docker for reproducible ML environments
 - Building, managing, and optimizing Docker images
 - **Kubernetes ecosystem:** pods, deployments, services, scaling
 - Deploying and managing ML workloads on Kubernetes clusters
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Unit 3 – Data Pipelines, APIs & Cloud Deployment

- Building and orchestrating data pipelines (Apache Airflow/Luigi)
 - Scheduling and monitoring ETL workflows
 - RESTful API development (Django/FastAPI) with MVC, ORM, and security
 - Cloud platforms for ML (AWS SageMaker, Azure ML, GCP AI Platform)
 - Kubernetes on cloud for scalable ML services
 - CI/CD for ML, testing, monitoring, logging, A/B testing, cost optimization
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Unit 4 – Security, Best Practices & Industry Integration

- Security & compliance in MLOps (GDPR, ISO)
- Case studies from e-commerce, finance, healthcare
- Lab: End-to-end corporate-style ML project from data acquisition to Kubernetes deployment
- Capstone mini-project with CI/CD integration
- Industry mentor guest lecture
- Certification readiness (AWS Certified ML Specialty or equivalent)

Suggested Text/s:

1. **Designing Machine Learning Systems** – Chip Huyen
 - **Why:** This is one of the most practical books on building and deploying ML systems at scale. Covers ML lifecycle, MLOps, data pipelines, experimentation, deployment, and monitoring in industry settings.
 - **Industry Relevance:** Written from a product and engineering perspective, with case studies from leading tech companies.
2. **Kubernetes for Machine Learning** – Emmanuel Raj
 - **Why:** Hands-on approach to running, scaling, and managing ML workloads with Kubernetes. Includes examples with Docker, Kubeflow, and cloud platforms.
 - **Industry Relevance:** Perfect for bridging the gap between data science and production infrastructure.

Course structure:

1. 30% mid semester exam,
2. 30% viva(alternate Wednesday, starting from 20th),
3. 40% assignments/project